MIL-G-50246 (MU)
28 January 1969

MILITARY SPECIFICATION

GRENADE, HAND, FRAGMENTATION, M61
METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for one type of fragmentation hand grenade designated as the M61.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1330

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DRAWINGS

U. S. ARMY MUNITIONS COMMAND

9241644 - Grenade, Hand, Fragmentation, M61, Metal Parts Assembly

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 9241644 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts.- The parts shall comply with all requirements specified in Drawing (dwg.) 9241644 and with all requirements specified in applicable specifications.

3.3 Workmanship.- All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, chips, scale, dirt, grease, corrosion products, salt deposits and other foreign material. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material.

3.4 First article inspection.- This specification makes provisions for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 First article inspection

4.2.1 Submission.- The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of twenty-five (25) metal parts assemblies and twenty-five (25) complete sets of parts (i.e. 25 each of every component and every sub-assembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including certificates of conformance for all materials and statements of findings showing results obtained for all tests conducted to determine compliance with this specification and specifications for plating. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.- Assemblies and components will be subjected by the Government to any or all of the examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.- If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements.

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4.3 Inspection provisions

4.3.1 Lot formation.- The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.- The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Closing cap assembly (see dwg. 8840282 covering a detail of dwg. 9241644).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Inside diameter of closing cap minimum (min.)	Gage	01001
102.	Outside diameter of base disc, maximum (max.)	Gage	01002
103.	Height of crimping lip, max.	Gage	01003
104.	Distance between base disc diameter and inside diameter of closing cap, min.	Gage	01004
Minor:	AQL 1.50 percent		
201.	Protective coating of interior missing or contains bare spot	Visual	01005
202.	Protective coating of exterior missing or contains bare spot	Visual	01006
203.	Base disc loose	Manual	01007
204.	Evidence of poor workmanship (see 3.3)	Visual	01008

4.3.2.2 Coil, spiral (see dwg. 8840279 covering a detail of dwg. 9241644).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Outside diameter of spiral coil compressed	Gage	02001
102.	Height of spiral coil compressed	Gage	02002

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Minor:	AQL 0.65 percent		
201.	Protective finish missing or base metal exposed.....	Visual	02005

4.3.2.3 Well (see dwg. 9212183 covering a detail of dwg. 9241644).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Concentricity of large and small inside diameter with thread.....	Gage	03001
102.	Small inside diameter, min.....	Gage	03002
103.	Large inside diameter, min.....	Gage	03003
104.	Depth of threaded cavity, min...	Gage	03004
105.	Major diameter of thread, min...	Gage	03005
106.	Minor diameter of thread, max...	Gage	03006
107.	Length to top of taper.....	Gage	03007
108.	Length to bottom, min.....	Gage	03008
109.	Thickness through bottom, min...	Gage	03009
110.	Thickness through side wall, min.....	Gage	03010
Minor:	AQL 1.50 percent		
201.	Small outside diameter, max.....	Gage	03011
202.	Outside diameter of threaded flange, min.....	Gage	03012
203.	Evidence of poor workmanship (see 3.3).....	Visual	03013

4.3.2.4 Assembly (see dwg. 9241644).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 1.00 percent		
101.	Concentricity of well with retaining ring	Gage	04001
102.	Diameter of flange of lower retaining ring, max	Gage	04002
103.	Length from crimping flange to bottom of well, min	Gage	04003
104.	Lock seam not crimped 360 degrees	Visual	04004
105.	Gap between lock seam and body	Visual	04005
106.	Well loose	Manual	04006
Minor:	AQL 4.00 percent		
201.	Total length, max	Gage	04007
202.	Outside diameter over lock seam.	Gage	04008
203.	Length from retaining ring to top of lock seam	Gage	04009
204.	Retaining ring loose	Manual	04010
205.	Solder above flush with upper surface of well assembly	Visual	04011
206.	Protective coating missing or base metal exposed	Visual	04012
207.	Paint or flux on well interior	Visual	04013
208.	Evidence of poor workmanship (see 3.3)	Visual	04014

4.3.3 Testing

4.3.3.1 Internal air pressure test, metal parts assembly (see dwg. 9241644) - Major defect - Code No. 05001.- This test shall be conducted 100 percent. Any metal parts assembly which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.1.

4.3.3.2 Notched wire inspection

4.3.3.2.1 Spiral coil notch measurement (see dwg. 8840279) - Major defect - Code No. 02006.- At a maximum of one hour (production time per machine) intervals, one spiral coil will be tested as specified in 4.4.2. Any coil which fails to comply with the requirements specified herein shall be classed defective. A defective coil shall require the contractor to subject the coils

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manufactured during this time interval to the acceptance test plan described in MIL-STD-105 single sampling plan, Code Letter G with an AQL of 0.40 percent. Failure to pass this test plan rejects all coils produced during this time interval. These sample coils shall be tested as specified in 4.4.2.

4.3.3.2.2 Wire coil notch measurement (see dwg. 8840279) - Major defect - Code No. 02007.- In lieu of the sampling associated with the spiral coil notch measurement specified in 4.3.3.2.1, when the wire is notched and recoiled, subsequent to spiral coil formation, two (2) samples of wire, each equivalent in length to the circumference of the notching roll shall be selected from each coil of wire. One sample shall be taken from the beginning of the coil and the other sample from the end. The first sample shall be dimensionally inspected to qualify the tool for use in production. The second sample shall be dimensionally inspected as a control of the tool during production.

If the qualification sample of the notched wire does not conform to the dimensions as prescribed on the applicable drawing, the tool used for the notching shall not be used in production.

If the control sample of the notched wire does not conform to the dimensions as prescribed on the applicable drawing, the tool used for notching the wire shall be removed from production. Further, that portion of wire produced since the last tool control check shall be returned to the contractor for corrective action.

In the event of adjustments made on the notching tool, or the introduction of a new tool, or the return of a reworked tool, the tool must pass the qualification test described above. When inspection is performed as specified in this paragraph, the AQL's listed in 4.3.3.2.1 will not apply.

4.3.4 Inspection equipment.- Equipment Tabulation Number ET - 9241644 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production

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and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Internal air pressure test of metal parts assembly.- Prior to painting, the metal parts assembly, less closing cap, shall be subjected to the internal air pressure test using test equipment specified in 4.3.4.

4.4.2 Notch measurement, Spiral Coil.- The Spiral Coil shall be unwound, in the area of largest diameter, to a length equal to the circumference of the notching wheel or thirty-six inches, whichever is the larger. Twenty notches along this length shall be checked for depth as specified in the drawing. The coil will be classed defective if one or more of these notches is outside the allowable limit. The coil shall then be checked visually over its entire length for any missing notches. A total of 24 notches or any 2 consecutive notches missing shall classify the coil defective.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing

5.1.1 Level C.- The parts shall be packaged and packed as specified in MIL-STD-1169.

5.2 Marking.- Marking of all packing shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.- Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

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c. Provisions for submission of first article samples.
d. Specification MIL-C-50245 (MU), latest issue, is applicable for procurement of Clip, Safety, for Hand Grenades - Dwg. 9219914.

6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).- Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

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6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.- All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise

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specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1330-A-201

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
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ATTN: SMUPA-ND

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-G-50246(MU)

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD [illegible] 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-0279

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